

Eye Instruments

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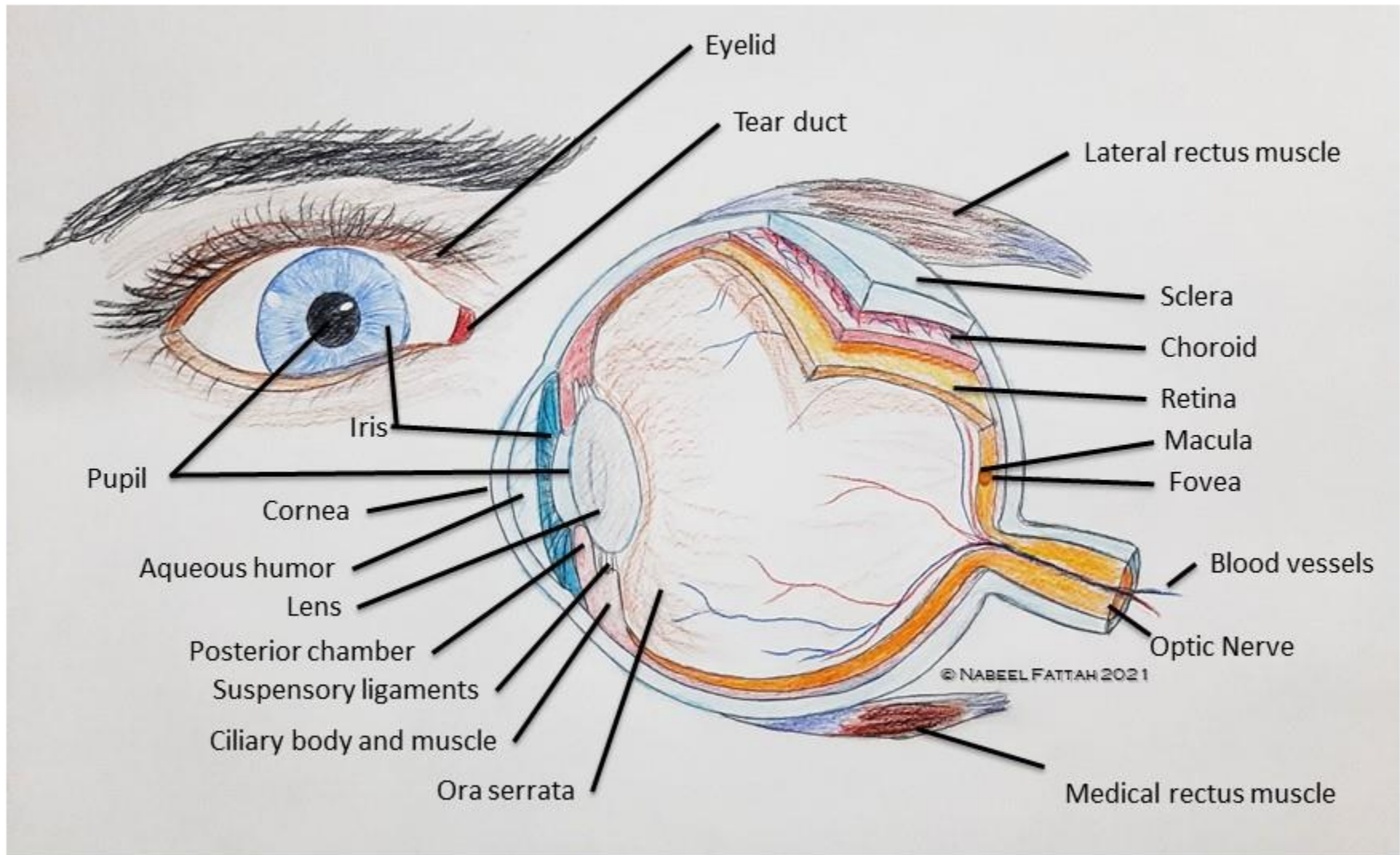
PhD Biomedical Engineer – England

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Diploma in Nursing and Patient Care - Ireland



Eye anatomy



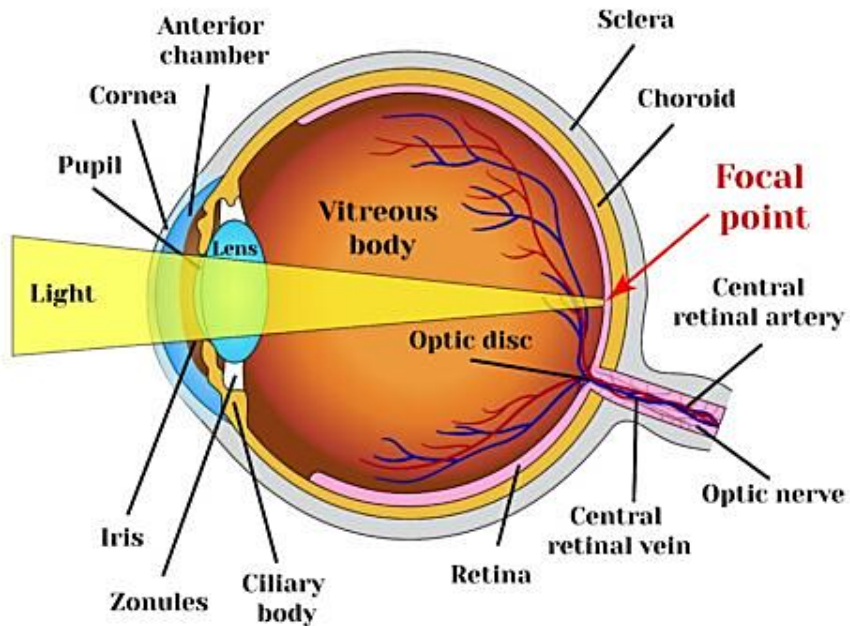
Illustrated by Nabeel Fattah 2021

Near point and far point

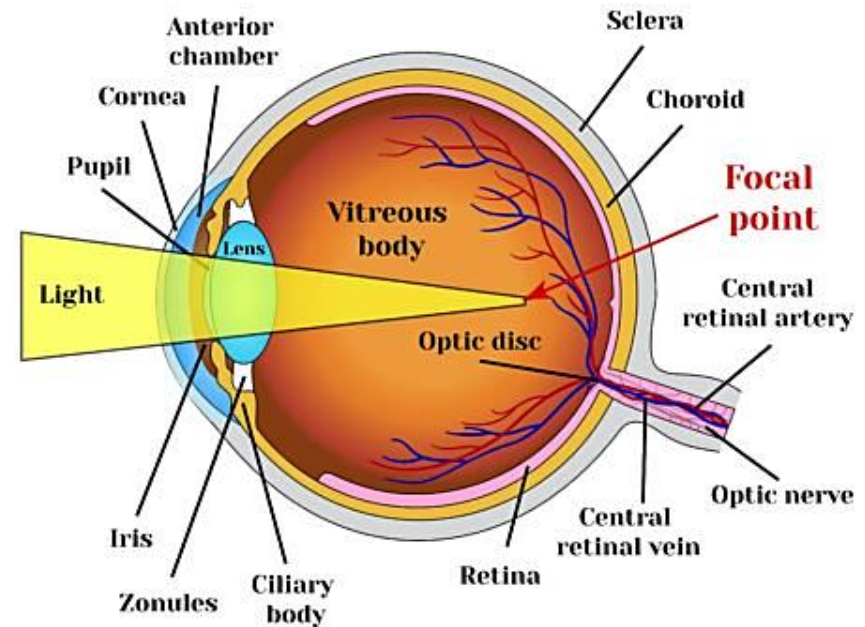
- The point nearest the eye at which an object can be placed and still produce a sharp image on the retina is called the near point (~ 25 cm from the eye- 20 yr old, 500 cm at age 60)

- The far point of the eye is the location of the farthest object on which a fully relaxed eye can focus. Normal vision people have a far point of nearly infinity

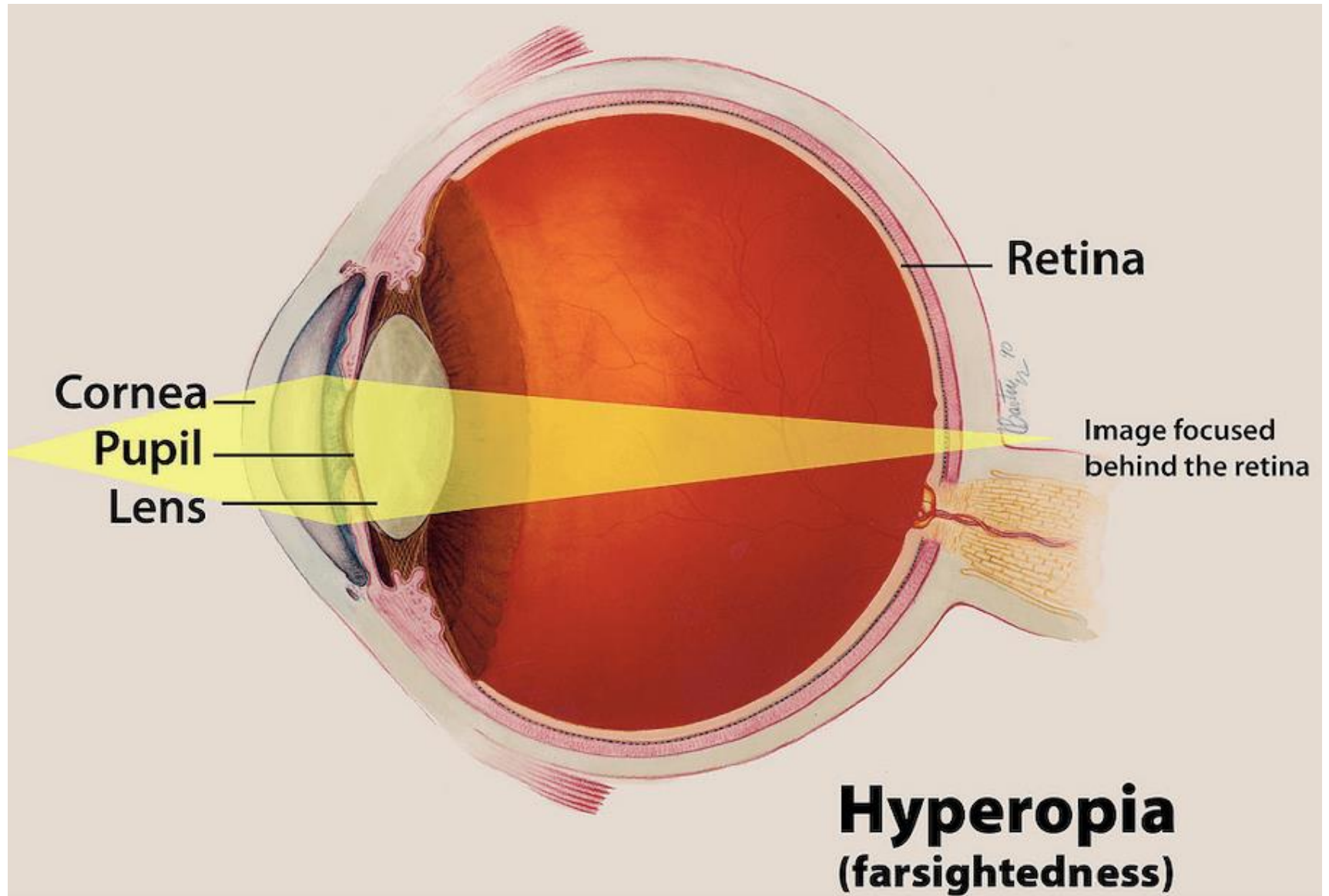
Normal vision



Nearsightedness Myopia



The process in which the lens changes its focal length to focus on objects at different distances is called **accommodation**





NORMAL VISION

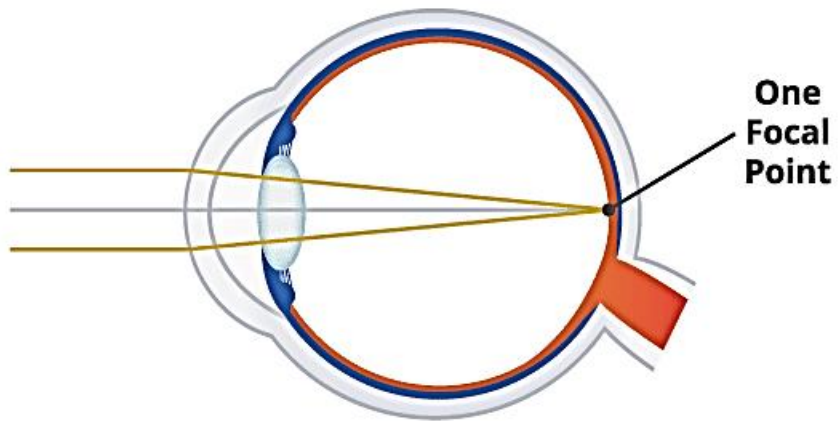


MYOPIC VISION

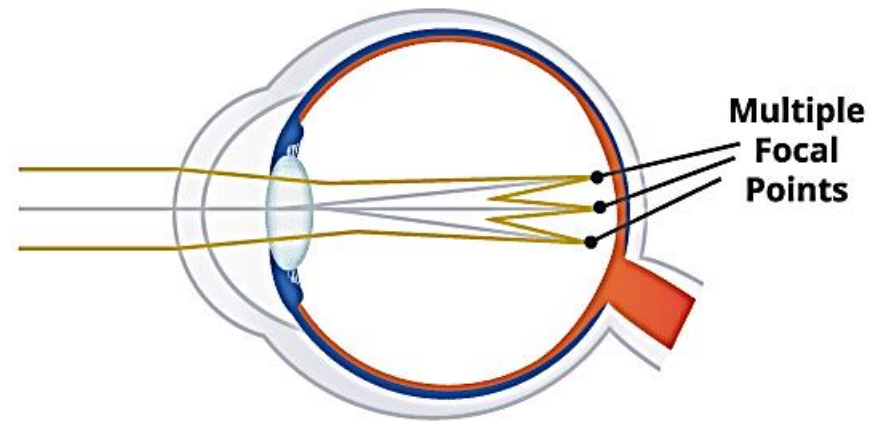


HYPEROPIC VISION

Normal Vision



Astigmatism

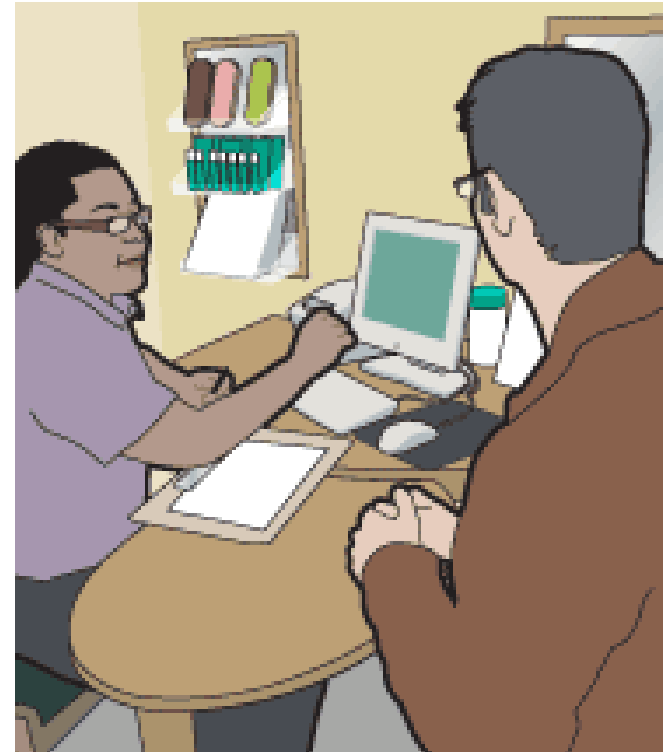


Guide for eye examination

- Regular eye examinations are essential for clear, comfortable vision.
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- They are an important health check too - the **optometrist** checks the health of your eyes and can find indications of other medical conditions.
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- There are hundreds of different ways to do the tests in your eye examination. This guide shows a typical example of each.

Visiting the Opticians

The optical assistant makes sure that your confidential customer record is accurate and up-to-date. If you are wearing contact lenses, you need to remove them before your eyes are tested.



Using a non-contact tonometer

Using a non-contact tonometer, the optical assistant (or the optometrist) blows a few puffs of air at each of your eyes in turn. The air bounces back at the instrument, giving a measurement of the pressure inside each eye. This is an important test, as high pressure can indicate the early stages of *glaucoma*, a sight-threatening condition.



The autorefractor

The autorefractor takes an electronic measurement of how well your eyes focus and gives a readout of your approximate prescription for the optometrist to use. If we do not have a record of your current prescription, the focimeter can read it from your glasses so that it can be compared with the findings of your eye examination by the optometrist.



The retinoscope

The optometrist may use an instrument called a retinoscope, which bounces a light beam off the back of your eye and back into the instrument. Different lenses focus the reflected light beam until it is steady, giving a close guide to the prescription you need. The retinoscope is very accurate - it is used to test the sight of very small children, or people with communication difficulties who can't easily describe how clearly they can see.



Using the ophthalmoscope

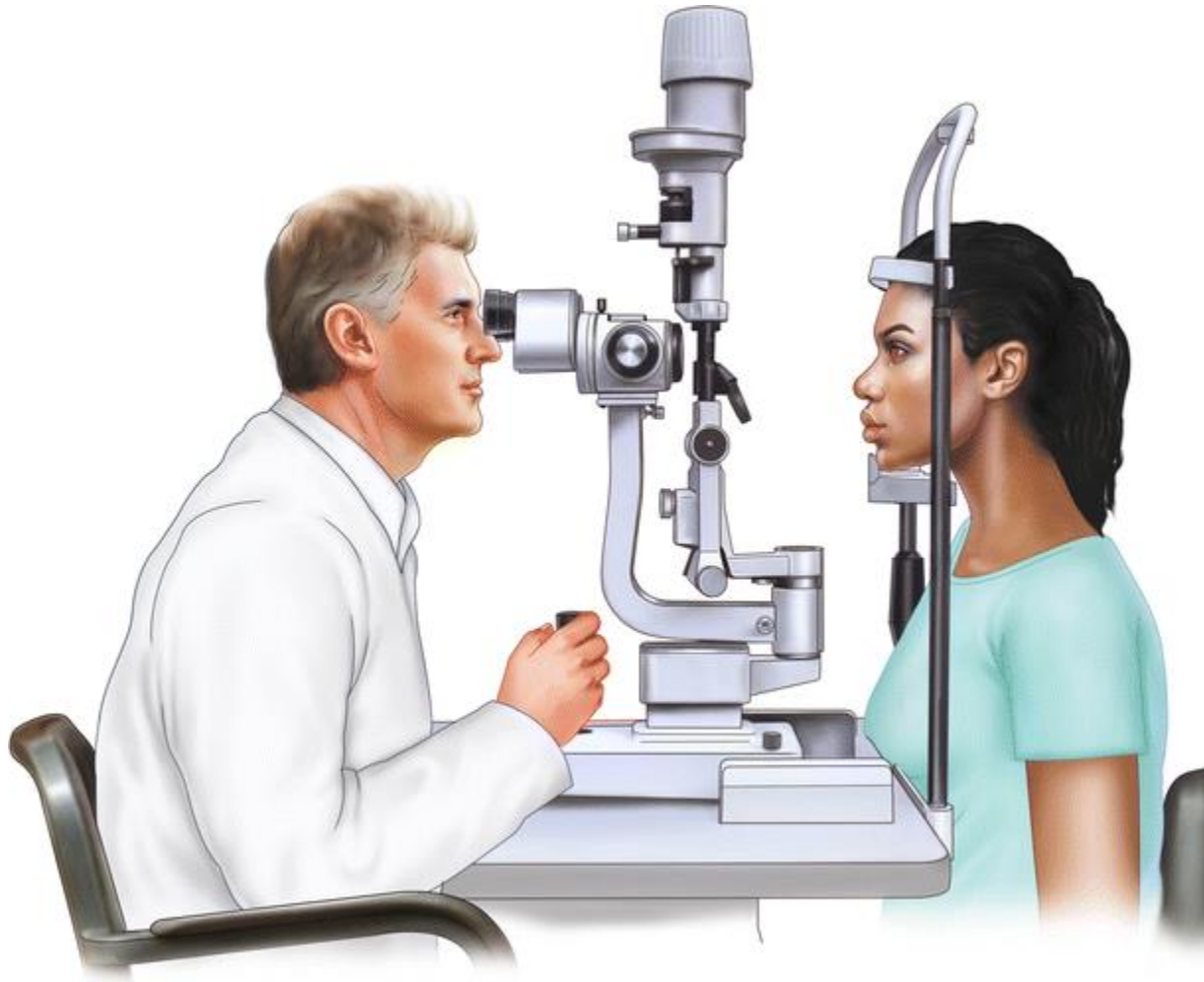
The optometrist uses an ophthalmoscope to examine the retina at the back of the eye, including the blood vessels and the front of the optic nerve. This important test can detect changes which can indicate diseases such as *diabetes* or *high blood pressure*.

The optometrist darkens the room and sits quite close to you, while he shines a bright light into each eye in turn using the ophthalmoscope. The light may leave shadows on your vision, but these soon fade.



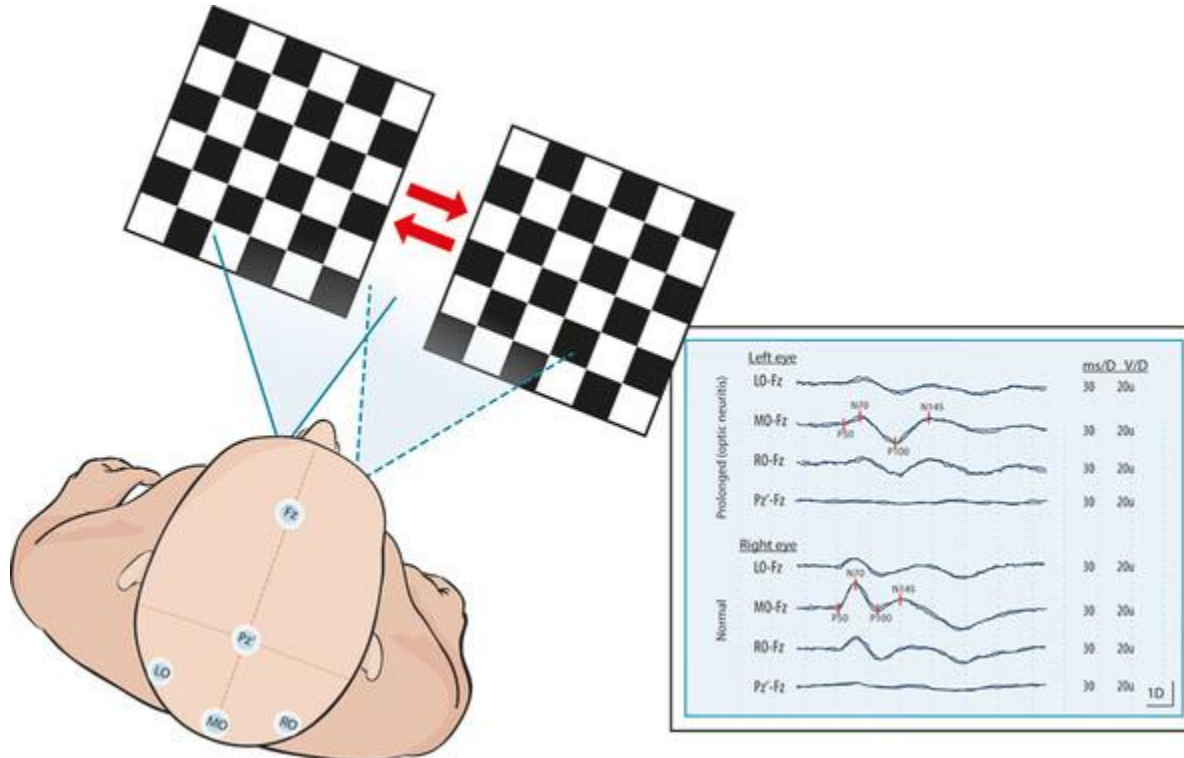
The slit lamp

The slit lamp is a powerful, illuminated microscope that is used to examine the outer surface of your eyes - the cornea, the iris and the lens - to check for *abnormalities* or *scratches*. This is a very important test for contact lens wearers.



Visual Evoked Potentials (VEP) Testing

Visual evoked potential is a test that records how the visual system records light. VEP tests are used to evaluate optic neuritis, optic tumours, retinal disorders, and demyelinating diseases such as multiple sclerosis.



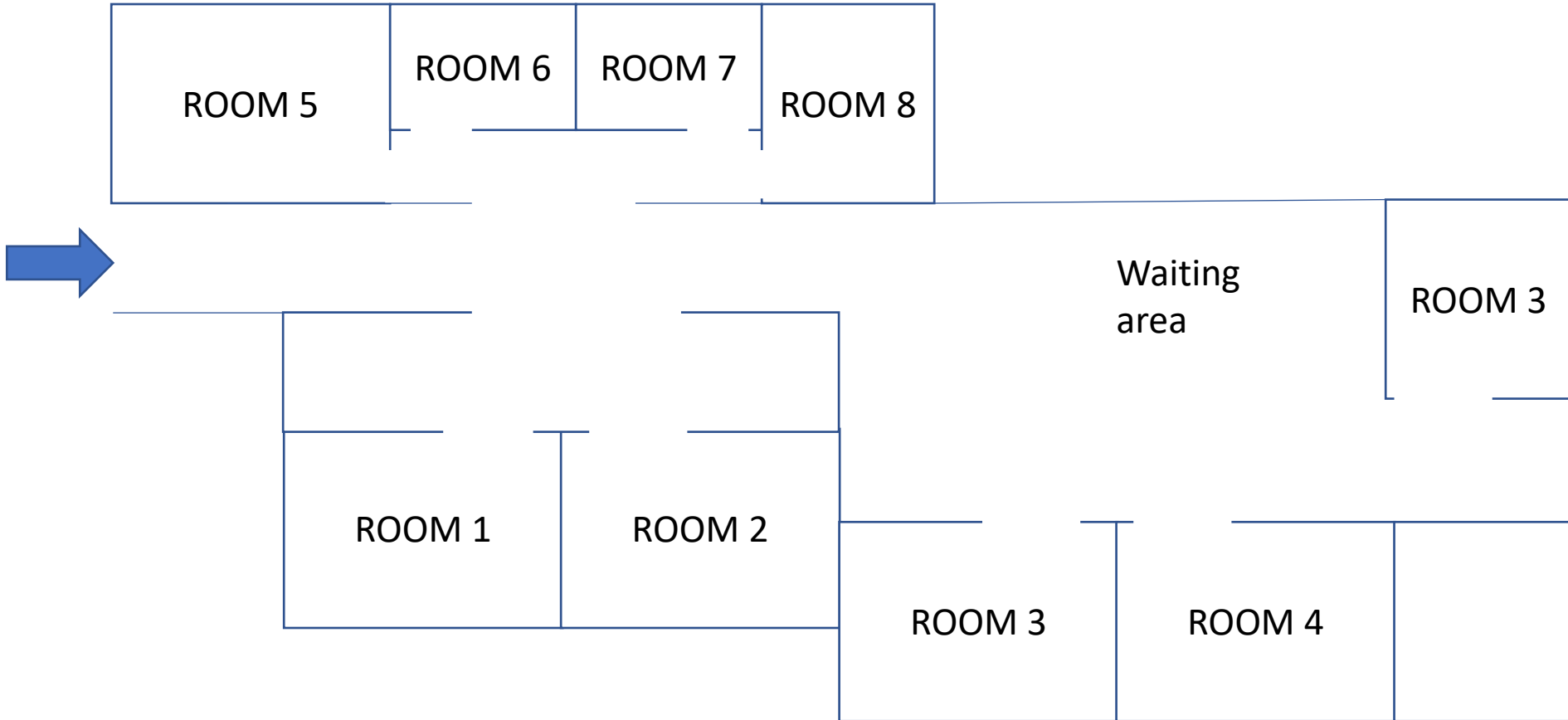
Electro-oculography

Electrooculography (EOG) is a test used for measuring the resting potential of the retina.

The EOG is used to assess the pigment epithelium of the retina and is useful in diagnosis Best's disease.



Eye consultant dept.



Task

In each room, note the names of the eye instruments.
You should be aware of each instrument's intended use.

Snellen chart is having different number of letters in different rows and the letter at top line should be read clearly at distance of 60 m. similarly the letters at subsequent lines as are read at 36, 24, 18, 12, 9, 6, 5mts respectively

E	1	20/200
F P	2	20/100
T O Z	3	20/70
L P E D	4	20/50
P E C F D	5	20/40
E D F C Z P	6	20/30
F E L O P Z D	7	20/25
D E F P O T E C	8	20/20
L E F O D P C T	9	
F D F L T C H O	10	
F E R O L C F T S	11	

Download Snellen chart

Download the chart from:

<http://retina-amd.org/wp-content/uploads/2017/10/SNELLEN-TEST.pdf>

Or from

https://www.provisu.ch/images/PDF/Snellenchart_en.pdf

Note: please follow the instruction for how to print it on the right way.

Case 1

Full name:

Age:

Gender:

R_eye:

L_eye:

Case 2

Full name:

Age:

Gender:

R_eye:

L_eye:

Case 3

Full name:

Age:

Gender:

R_eye:

L_eye:

Case 4

Full name:

Age:

Gender:

R_eye:

L_eye: